

The Role of Open Source in Building Modern Data Infrastructure

It's no secret that open source is emerging as the backbone of modern data infrastructure. Here's a list of the core open source technologies used to deploy this infrastructure, along with some real-world examples and a brief on why open source matters.



In the era of Big Data, artificial intelligence, and cloud-native development, modern data infrastructure is the foundation that enables organisations to derive insights, automate decisions, and power digital products. With the volume, variety, and velocity of data growing at unprecedented rates, enterprises need scalable, flexible, and efficient systems. Open source technologies have emerged as a cornerstone of modern data infrastructure, driving innovation, reducing costs, and providing the flexibility that proprietary software often lacks.

What is modern data infrastructure?

Modern data infrastructure is an integrated architecture of tools, platforms, and processes that manage the entire lifecycle of data—from ingestion and storage to processing, analysis, and visualisation. It typically includes:

- Data ingestion systems (e.g., Kafka, Flume)
- Data storage solutions (e.g., Hadoop HDFS, Apache Iceberg, Delta Lake)
- Data processing engines (e.g., Apache Spark, Flink)
- Data orchestration tools (e.g., Apache Airflow, Dagster)
- Data transformation and modelling frameworks (e.g., dbt, Apache Beam)
- Query engines (e.g., Presto, Trino, Druid)
- Monitoring and observability (e.g., Prometheus, Grafana)
- Data governance and metadata management (e.g., Amundsen, Apache Atlas)

Most of these tools are open source and form the backbone of modern data ecosystems.

Why open source matters in data infrastructure

Cost-effectiveness and accessibility: Open source tools